

Sales & Profit Analysis

End-to-end case study: Python, SQL, Excel and Power BI

Week 4 capstone, Data Analyst course (Skill Nexus) | Prepared by Alekya | October 2026

118.73M	16.89M	14.2%	1.13M	58
Total sales	Total profit	Profit margin	Units sold	Loss-making orders

Executive summary

This report analyses 700 sales orders from September 2013 to December 2014 across five countries, five customer segments and six products. The business earned 118.73M in sales and 16.89M in profit (a 14.2% margin). Profit is highly concentrated, and one segment loses money.

- **Government drives the business.** It is 44.2% of sales but 67.4% of profit (11.39M profit, 21.7% margin).
- **Enterprise is the only loss-making segment** (-0.61M, -3.1% margin). All 58 loss-making orders in the data are Enterprise orders, and every Enterprise order at a Medium or High discount lost money.
- **Discounts erode margin.** Margin falls from 21.9% with no discount to 9.1% in the High band. High-discount orders are 31.5% of sales but only 20.1% of profit.
- **The United States sells the most but earns the least per dollar.** It has the highest sales (25.03M) and the lowest margin (12.0%), and the largest Enterprise loss (-175K).
- **Like-for-like growth is strong.** Sep-Dec 2014 sales were 36.9% higher than Sep-Dec 2013, and profit was 40.1% higher.

The main recommendation is to cap Enterprise and High-band discounts. Section 7 lists six actions with the evidence behind each.

1. Objective

Identify which segments, countries, products and discount levels drive profit, and where the business is losing money, so that a stakeholder can decide where to focus pricing and sales effort.

2. Dataset

700 orders and 16 columns: Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month Number, Month Name and Year. Period: September 2013 to December 2014. Countries: Canada, France, Germany, Mexico and the United States. Segments: Government, Small Business, Enterprise, Midmarket and Channel Partners. Products: Paseo, VTT, Amarilla, Velo, Montana and Carretera.

3. Tools and method

Step	Tool	What was done
Clean and explore	Python (Pandas, NumPy), Google Colab	Loaded, inspected, validated and cleaned the data; added Profit_Margin_% and Year_Month.
Analyse	Pandas and SQL (SQLite)	Profit, sales and margin by segment, country, product, discount band and month; SQL queries on discounts and loss-making orders.
Visualise	Excel, Power BI Desktop, Matplotlib/Seaborn	Pivot tables with slicers in Excel, a Power BI dashboard, and Python charts.
Report	This document	Findings, recommendations and limitations.

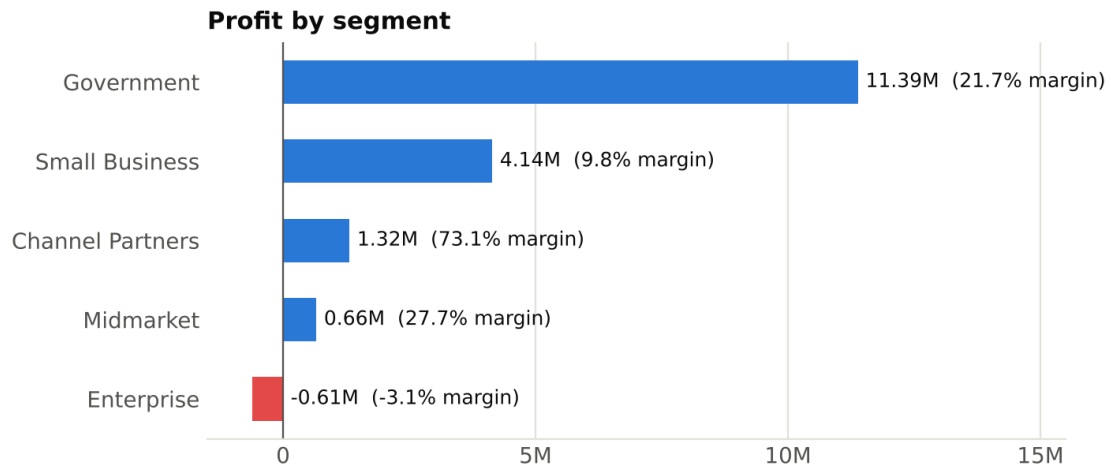
4. Data cleaning and quality checks

Check	Result
Size	700 rows x 16 columns
Missing values	None found
Duplicate rows	None found, so no rows were removed
Dates	All 700 converted to real dates; the month matches Month Number in every row
Sales = Gross Sales - Discounts	0 mismatches
Profit = Sales - COGS	0 mismatches
Discount band "None"	Kept as a real category (it means no discount), not treated as missing data
Column names	Trimmed and standardised (the Sales column had a leading space); spaces replaced by underscores
Derived columns	Profit_Margin_% (Profit / Sales) and Year_Month

The totals tie out: the Excel pivots, the Power BI dashboard (118.73M sales, 16.89M profit, 1.13M units) and the Python summary tables all agree.

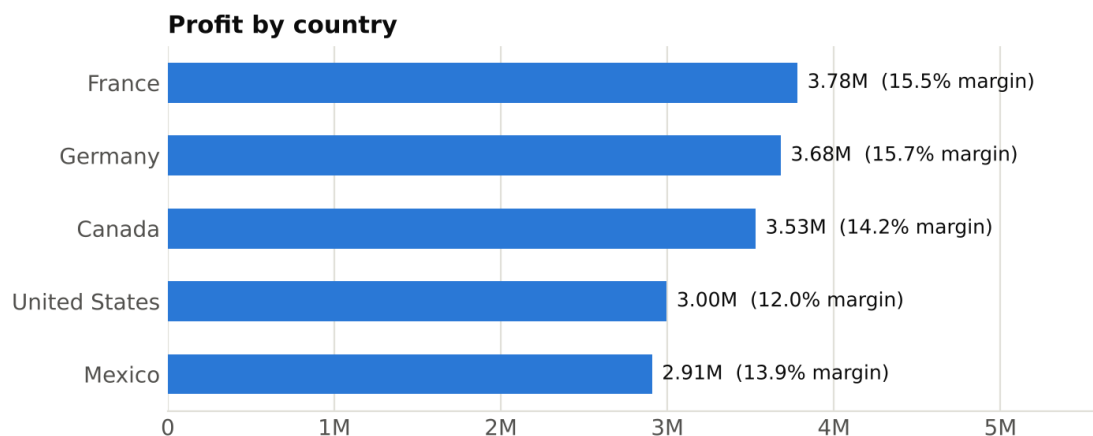
5. Findings

5.1 Segment



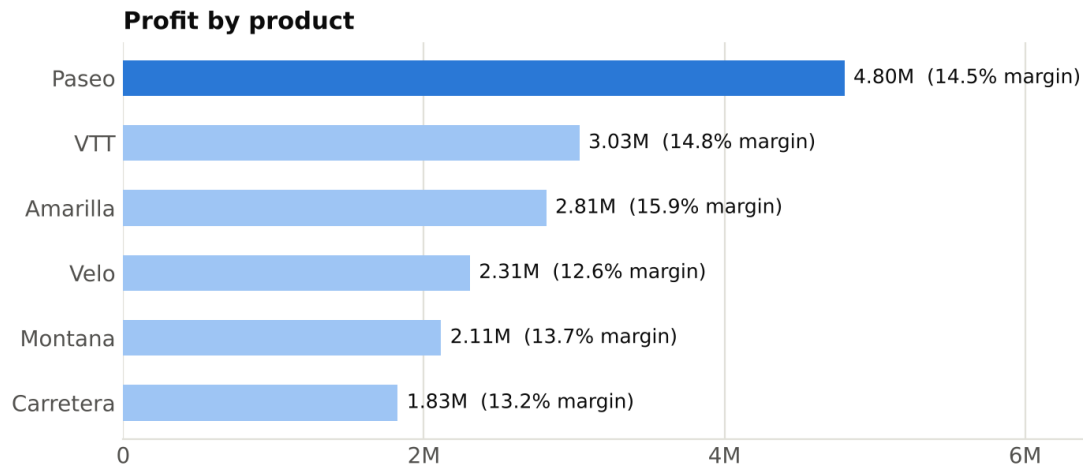
- **Government** generates 52.50M of sales and 11.39M of profit. It is the largest and most reliable profit source in every country.
- **Small Business** has 42.43M of sales but only 4.14M of profit (9.8% margin), because COGS takes about 90% of its sales.
- **Channel Partners** (73.1% margin) and **Midmarket** (27.7%) have the best margins, but together they are only 3.5% of sales and 11.7% of profit. Their unit prices are low (12 and 15 per unit), so the high margin sits on a small base.
- **Enterprise** has 19.61M of sales but COGS equals 103% of sales, giving a net loss of 0.61M. It loses money in all five countries.

5.2 Country



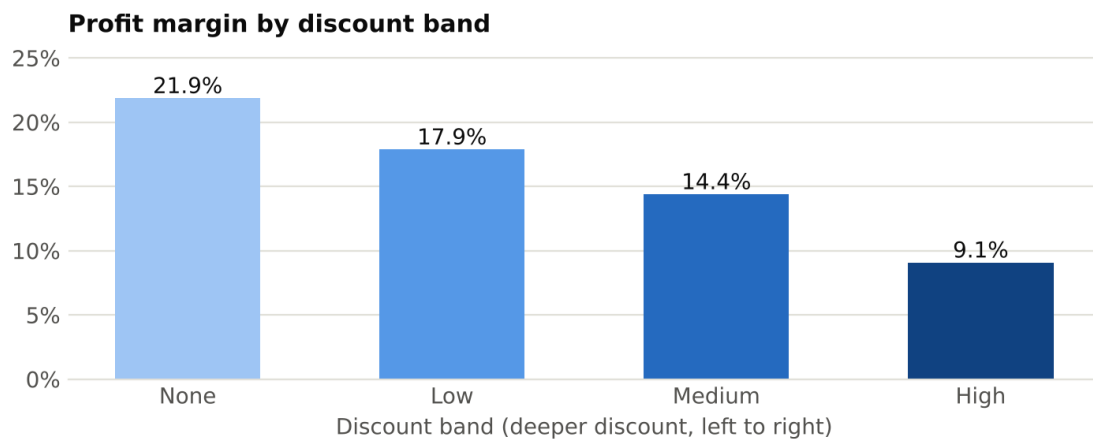
- France, Germany and Canada lead on profit (3.78M, 3.68M, 3.53M).
- The **United States** has the highest sales (25.03M) but the lowest margin (12.0%) and the largest Enterprise loss (-175K, against -96K to -122K elsewhere).
- **Mexico** has the lowest sales (20.95M) and the lowest profit (2.91M).

5.3 Product



- **Paseo** is the core product: 33.01M of sales and 4.80M of profit, which is 28.4% of all profit.
- Margins across products are narrow (12.6% for Velo to 15.9% for Amarilla), so product mix explains much less than segment and discount do.
- **Carretera** has the lowest profit (1.83M) and the largest total loss on loss-making orders (-231K).

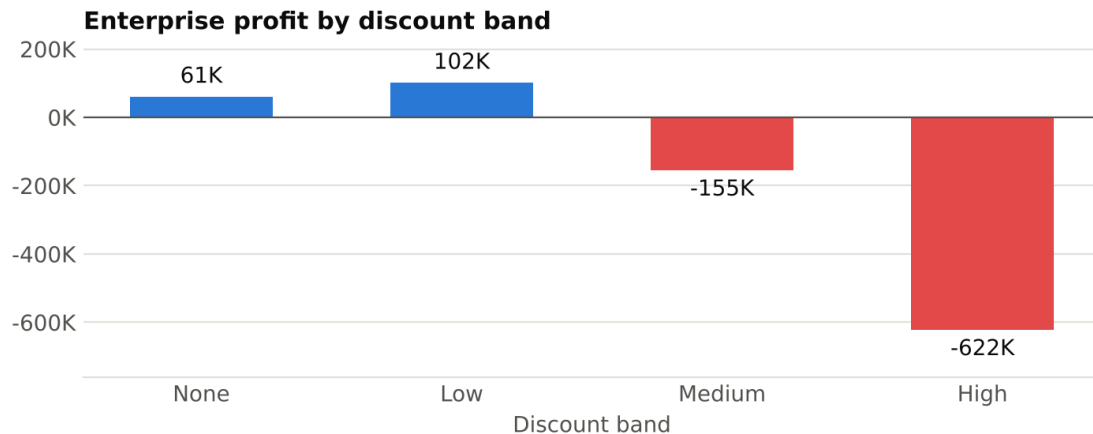
5.4 Discounts



Discount band	Orders	Sales	Profit	Margin	Avg profit per order
None	53	7.94M	1.74M	21.9%	32,763
Low	160	34.63M	6.19M	17.9%	38,680
Medium	242	38.78M	5.58M	14.4%	23,056
High	245	37.37M	3.39M	9.1%	13,832

Margin falls at every step as the discount deepens. The High band holds 35% of orders and 31.5% of sales but only 20.1% of profit. This is an observed association in the data, not proof that discounts alone cause the lower margin.

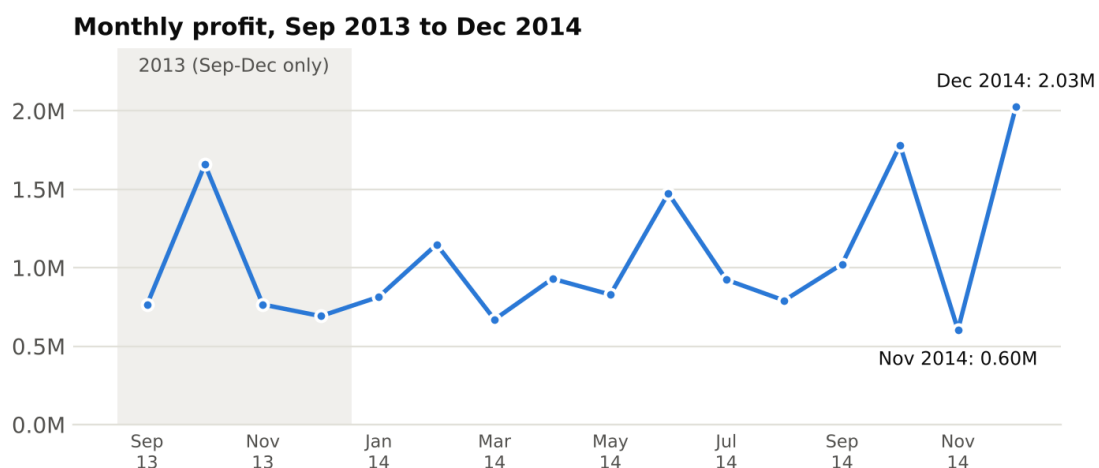
5.5 Why Enterprise loses money



- Enterprise lists at 125 per unit and costs 120 per unit (COGS divided by units sold). Any discount above about 4% therefore makes an order loss-making.
- The average Enterprise discount is 6.9%, so the realised price (about 116) is below cost.
- All 58 Enterprise orders in the Medium and High bands lost money (-777K in total). The 42 orders with no or Low discount earned +163K.

The unit price and unit cost above are derived from the dataset (Sale Price and COGS divided by Units Sold).

5.6 Trend over time



- Profit peaked in December 2014 (2.03M) and dipped in November 2014 (0.60M). October was strong in both years (1.66M in 2013, 1.78M in 2014).
- **Read year totals with care.** The data starts in September 2013, so 2013 has only four months (175 of 700 orders). The full-year gap (26.4M sales in 2013 versus 92.3M in 2014) overstates growth.
- Comparing the same four months (Sep-Dec) gives sales growth of 36.9% (26.4M to 36.2M) and profit growth of 40.1% (3.88M to 5.44M). Margin was stable at 14.7% in 2013 and 14.1% in 2014.

6. Dashboard

The Power BI dashboard shows the headline KPIs, sales and profit by year, and profit by segment, country and product. The Excel workbook has matching pivot tables and charts with Country and Year slicers.

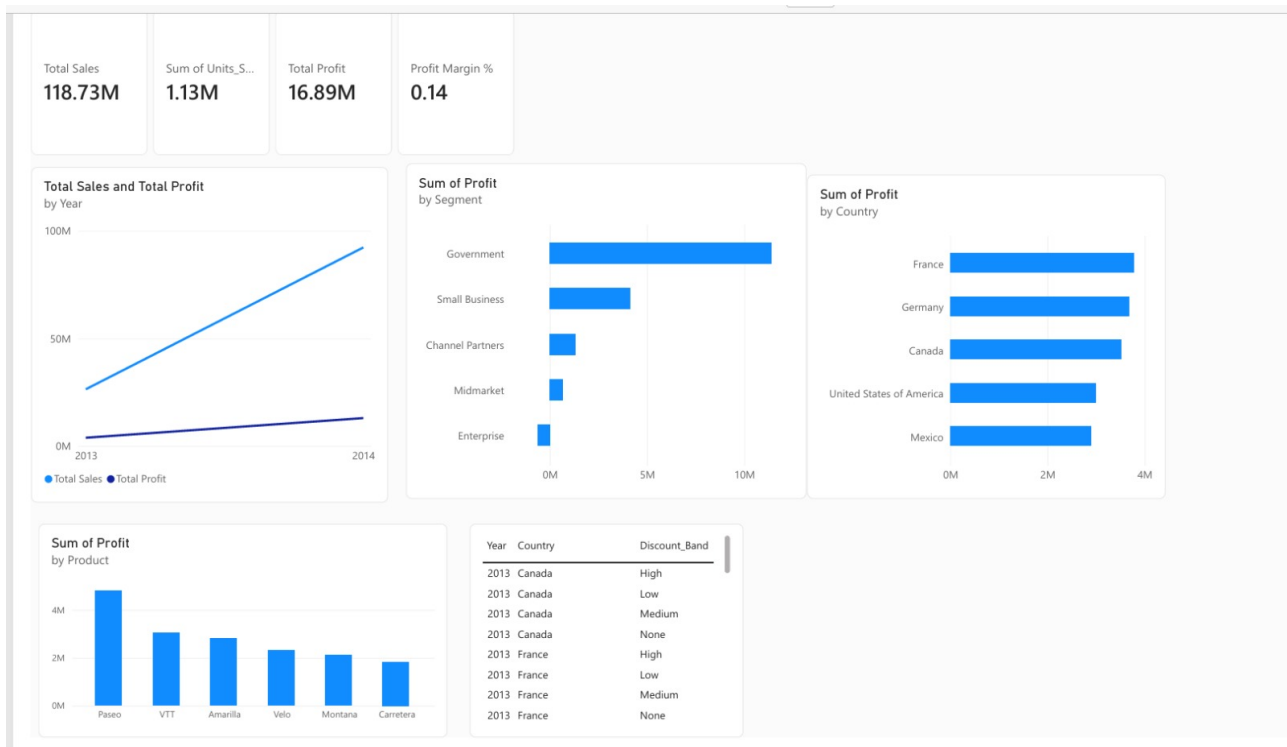


Figure: Power BI dashboard with KPI cards, sales and profit by year, and profit by segment, country and product.

7. Recommendations

#	Recommendation	Evidence	Indicative effect
1	Stop deep discounting in Enterprise. Cap discounts at the Low band (break-even is about 4% at today's prices), or raise the list price toward about 129 so the average 6.9% discount breaks even.	All 58 loss-making orders are Enterprise orders in the Medium and High bands.	Up to +0.78M profit (4.6% of current profit) if those losses are removed.
2	Tighten High-band discounts across all segments. Require approval above a set threshold.	High band margin is 9.1% against 14.4% for Medium; it is 31.5% of sales and 20.1% of profit.	About +1.99M if High-band orders earned the Medium margin (illustrative; overlaps with #1; assumes volume is unchanged).
3	Protect Government and watch concentration risk. Keep pricing discipline and diversify over time.	67.4% of profit comes from one segment.	Protects 11.39M of profit.
4	Review Small Business costs and pricing.	35.7% of sales but only 9.8% margin; COGS is about 90% of sales.	Each +1 point of margin is about +0.42M profit.
5	Investigate US pricing and discounts.	Highest sales, lowest margin (12.0%), largest Enterprise loss (-175K).	Closing half the gap to Germany's 15.7% margin would add about 0.46M.
6	Test growth in Channel Partners and Midmarket before shifting resources.	Margins of 73.1% and 27.7%, but only 3.5% of sales and low unit prices.	Upside is real but unproven at scale.

Effects are simple what-if estimates from this dataset and assume order volumes stay the same. Test them before committing.

8. Limitations

- 2013 covers only September to December, so year-on-year comparisons must be made on the same months.
- The data is a 700-order sample with no customer or order IDs. The analysis is descriptive, so associations (for example, discount and margin) are not proof of cause.
- Channel Partners' 73.1% margin rests on 100 low-price orders and should not be extrapolated.
- Cost detail is limited to COGS, so unit price and unit cost figures are derived rather than recorded.

9. Conclusion

The business is profitable (14.2% margin) but depends on Government for two-thirds of its profit. The clearest fix is discount discipline: Enterprise orders discounted beyond about 4% lose money, and deep discounts cut margin across the whole business. Tightening discounts, reviewing Small Business and US pricing, and testing growth in the high-margin small segments give a practical plan with evidence behind each step.

Appendix: project files

Notebook (sales_analysis.ipynb), cleaned data (cleaned_sales_data.csv), charts folder, analysis_summary.xlsx, excel_pivot_analysis.xlsx, sales_dashboard.pbix and this report. All files are in the GitHub repository sales-profit-analysis.